

Bridging Composition and Transfer Operators with Locally Linear and Diffusive Kernels

Patrick S. Mahon

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Abstract

Across non-linear dynamics, machine learning, and statistical physics, two operator viewpoints are prevalent: the Koopman perspective, which treats dynamics as deterministic composition on observables, and the Perron–Frobenius (transfer) perspective, which treats evolution as the spreading of densities. Each excel while leaving an interpretive gap between ‘pure chaos’ and ‘pure noise’.

We propose a bridge: a data-driven, two-dialed operator family. One dial controls how local the estimated drift is; the other controls how strongly residual variability is interpreted as diffusion. Sweeping the dials traces a continuous surface whose edges recover the Koopman and Perron–Frobenius pictures and whose interior supplies every intermediate mixture.

Framing this surface as a homotopy in operator space clarifies where familiar techniques—delay embeddings, diffusion maps, local-linear forecasting, stochastic model reduction—sit inside the same conceptual square. Practical diagnostics based on residual spectra reveal the minimal embedding dimension and quantify when unresolved curvature is better modeled as stochastic forcing. A continuous-time limit connects the Rose construction to classical Itô SDEs and their Fokker–Planck and Kolmogorov equations, providing a link between the data scientific and stochastic analysis perspectives.

Our goal is not to champion yet another single algorithm or approach, but provide a common scaffold. The framework helps practitioners express where they stand on the spectrum of operator models and how to move, continuously and transparently, to the representation that best fits their data and scientific questions.

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1 Local Linearizations

Throughout Sections §1–§4 the state space X will be a separable Banach space endowed with norm $\|\cdot\|_X$. Finite–dimensional $\mathbb{R}^d$ examples are recovered by taking $X = \mathbb{R}^d$ with its Euclidean norm, but the analytic statements below (Taylor expansion, Fréchet derivatives, Kolmogorov generators) require only Banach-space calculus; see [28] for a concise reference.

We begin with the Cauchy problem

$$\begin{aligned} \frac{du}{dt} + A(u(t)) &= 0, \quad t > 0, \\ u(0) &= u_0, \quad u_0 \in U \subset X, \end{aligned} \tag{1}$$

where A is an operator (possibly nonlinear and state-dependent) that generates a *contraction* C_0 -semigroup $T(h)$ on the Banach space $\mathcal{X}$.¹

In many practical settings, one only has access to discrete observations of the state. Let these observations be arranged in the data matrix

$$X_t = \begin{bmatrix} x(t) \\ x(t-h) \\ x(t-2h) \\ \vdots \end{bmatrix}, \tag{2}$$

where the rows correspond to states sampled at time intervals of length h .

¹Background on contraction semigroups and their generators can be found in [28, Chap. 1].

A first approach is to attempt a global linearization via a least-squares solution to,

$$X_t \approx X_{t-h} T(h), \quad (3)$$

$$\text{or equivalently, } u(t) \approx T(h) x(t-h). \quad (4)$$

56 If the residuals from this approximation are well within an acceptable and prescribed tolerance—then
it is reasonable to assume that the dynamics are well represented by a linear operator. In such a
58 case, one can recover the global generator via

$$A = \lim_{h \rightarrow 0^+} \frac{1}{h} (T(h) - I). \quad (5)$$

This approach is particularly effective when A is linear or nearly linear. The contraction property
60 aids in establishing the existence and uniqueness of solutions, however it also means that a global
linearization of A may be unattainable when its dependence on $u(t)$ is significant with residuals
62 remaining unacceptably large.

A global linearization may fail to capture the system’s behavior when A is state dependent. In such
64 case a local linearization may be preferable.

To construct a local approximation we can introduce a weighting operator W designed to select
66 or emphasize those rows in X_t that are dynamically relevant. The idea is to filter the data so
that regions exhibiting dynamics similar to those surrounding $x(t)$ are highlighted, typically by
68 considering observations local to $x(t)$ under some metrization of $\mathcal{X}$.

If W is constructed using a localization kernel $k_\theta(x(j), x(i))$, where θ parametrizes what “local”
70 means, then the weighted observations take the familiar *locally-weighted least-squares* form [2, 3,
8],

$$W_\theta X_t, \quad (6)$$

72 and the local approximation is expressed as

$$W_\theta X_t \approx W_\theta X_{t-h} \hat{T}(h, \theta, t), \quad (7)$$

where $\hat{T}(h, \theta, t)$ is the locally linearized evolution operator valid in a neighborhood of $x(t)$. If a
74 weighted pseudoinverse is employed,

$$A(x(t), \theta, h) = \left((W_\theta X_{t-h})^\dagger (W_\theta X_t) - I \right) \quad (8)$$

where $(W X_{t-h})^+$ denotes the pseudoinverse of $W X_{t-h}$. Now recovery of the generator $A(x(t))$
76 requires the spatial limit in θ , followed by the temporal limit in h ,

$$A(x(t)) = \lim_{h \rightarrow 0} \frac{1}{h} \left(\lim_{\theta \rightarrow 0} \hat{T}(h, \theta, t) - I \right) \quad (9)$$

If T is (i) deterministic (i.e. $\theta \rightarrow 0$ iff, in the weak sense, $k_\theta \rightarrow \delta$) and (ii) locally linear (so that
78 $T = \sum_{k=0}^{\infty} \frac{h^k}{k!} A(h, \theta, x_j)^k$ converges), then the framework recovers the Sugihara–May local linear
forecast paradigm [33] while remaining fully non-parametric.

80 However, in the absence of any strong *a priori* intuition regarding even the most general assumptions
of the generating process, most notably stochasticity and the rate of decay of higher order terms,
82 deciding how to approach these limits (and thus what constitutes a *good* linearization) is not
easy.

2 Composition and Transfer Operators in a Banach Setting

Throughout §2–§3.1 the phase space $(X, \|\cdot\|_X)$ is the separable Banach space fixed in §1. A finite, Borel reference measure μ endows X with the structure of a metric-measure space; all L^p -spaces, adjoints, and operator norms below are taken with respect to (X, μ) . (For background on measurable dynamics and Markov kernels in this setting see [18, 20].)

All L^p -spaces, adjoints, and operator norms below are taken with respect to the pair (X, μ) . Where $f : X \rightarrow \mathbb{R}$ appears, tacitly assume $f \in L^2(X, \mu)$; where ρ appears, assume $\rho \in L^1(X, \mu)$.

Definition 2.1 (Composition (Koopman) operator). Let $T_h : X \rightarrow X$ be the time- h map of the flow (1). For any square-integrable observable $f \in L^2(X, \mu)$ define

$$(Kf)(x) = f(T_h x)$$

- K is linear but acts on an *infinite-dimensional* function space even when $\dim X < \infty$.
- If T_h preserves μ then K is unitary and its spectrum lies on the unit circle [19, 23].
- No smoothing occurs: higher-order nonlinearities of the flow are implicit in composition, so K captures sharp deterministic features but is blind to stochasticity.
- For the Dirac kernel $p_\delta(x' | x) = \delta(x' - T_h x)$, K is the *left* adjoint of the deterministic Perron–Frobenius operator P_δ (see (10) below).

Global linearization (3) corresponds to approximating K by a finite-rank operator on $L^2(X, \mu)$; this is precisely the operator-theoretic viewpoint exploited in modern “Koopman analysis” [1, 6].

Definition 2.2 (Transfer (Perron–Frobenius) operator). Let $p(x' | x)$ be a Markov transition density. For any $\rho \in L^1(X, \mu)$ set

$$(P\rho)(x') = \int_X \rho(x) p(x' | x) d\mu(x) \tag{10}$$

- P is positive and mass-preserving: $\int (P\rho) d\mu = \int \rho d\mu$ [20].
- Under mild regularity P is a strict contraction on the mean zero subspace $L_0^2(X)$ whenever p has non-zero width; its spectral radius is then < 1 (cf. [13, Eq. (10), p. 209]).
- A non-trivial width encodes diffusion, measurement error, or model misspecification. In local linearization the residual cloud $r_t^{(\theta)}$ naturally induces such a kernel.
- For the deterministic kernel $p_\delta(x' | x) = \delta(x' - T_h x)$, P_δ is the *right* adjoint of the Koopman operator (2.1).

Throughout we fix a reference measure μ ; all L^p spaces, inner products, and adjoints are taken with respect to μ .

Two extreme lenses. Koopman K captures deterministic curvature but ignores noise, whereas a non-delta transfer operator P absorbs all residual variation as diffusion. Global least-squares (3) implicitly demands that K admit an accurate finite-rank approximation; large residuals signal failure. Conversely, shunting all residual energy into P treats it as pure diffusion.

Need for a continuous bridge. We therefore seek a *continuous family* of operators

$$(\theta, \sigma) \mapsto (\mathcal{R}_{\theta, \sigma}^b, \mathcal{R}_{\theta, \sigma}^\sharp) \quad (11)$$

such that $\lim_{\theta \rightarrow 0, \sigma \rightarrow 0} \mathcal{R}_{\theta, \sigma}^b = K$ and $\lim_{\theta, \sigma \rightarrow \infty} \mathcal{R}_{\theta, \sigma}^\sharp = P_\infty$ (the maximally diffusive rank-one PF operator).

The construction is *local* (weighted neighbourhoods in X) and *data-adaptive* (empirical residual covariance). Together these yield a two-parameter homotopy surface: shrinking θ refines the deterministic drift operator, enlarging σ injects diffusion. All subsequent population-level statements (up to §3.2) hold verbatim for the Banach space just fixed; only in §?? do we specialize to manifolds and curvature.

3 From Koopman and PF to a Two-Dial Family

The preceding section placed the deterministic pull-back *Koopman* and the stochastic push-forward *Perron–Frobenius* operators at opposite ends of the modeling spectrum [19, 29, 12, 20]. Working still in the separable Banach space $(X, \|\cdot\|_X)$ we now build a two-parameter family of operators,

$$(\theta, \sigma) \mapsto (\mathcal{R}_{\theta, \sigma}^b, \mathcal{R}_{\theta, \sigma}^\sharp), \quad (\theta, \sigma) \in (0, \infty) \times (0, \infty)^2 \quad (12)$$

with the following properties:

1. $\lim_{\theta \rightarrow 0, \sigma \rightarrow 0} \mathcal{R}_{\theta, \sigma}^b = K$ and, dually, $\lim_{\theta \rightarrow 0, \sigma \rightarrow 0} \mathcal{R}_{\theta, \sigma}^\sharp = P_\delta$ (the deterministic PF operator with Dirac kernel).
2. $\lim_{\theta, \sigma \rightarrow \infty} \mathcal{R}_{\theta, \sigma}^\sharp = P_\infty$ where P_∞ is the rank-one transfer operator whose kernel is a spatially uniform density, and $\mathcal{R}_{\theta, \sigma}^b$ collapses observables to the corresponding constant.
3. The map $(\theta, \sigma) \mapsto (\mathcal{R}_{\theta, \sigma}^b, \mathcal{R}_{\theta, \sigma}^\sharp)$ is jointly continuous in the L^2 -operator norm; varying θ (locality) and σ (residual blur) in either order yields the same result, giving a *homotopy of homotopies*—a filled square in the 2-category $\text{End}(L^2(X, \mu))$.

We call this two-dial construction the **Rose operator family**. Conceptually, it unifies earlier local linear predictors [10, 33, 9] with kernel-smoothing views of stochastic transport [24, 35, 27, 30].

- a *locally weighted linear predictor* $C_x^{(\theta)}$ that refines deterministic drift as $\theta \downarrow 0$, and
- an *empirical residual kernel* governed by g_σ that injects increasing diffusion as $\sigma \uparrow \infty$.

Because the θ - and σ -axes commute, one may independently *sharpen drift* (decrease θ) or *blur noise* (increase σ), tracing paths across the square that suit the data at hand. In the interior of this “Rose surface” lie operator pairs that blend deterministic curvature with state-dependent or homogeneous diffusion—precisely the regime where, presumably, most representations arise.

² $(0, \infty) \simeq [0, 1)$ via $u \mapsto \frac{1}{1+u}$; we keep the positive radii notation for engineering clarity.

3.1 Two-parameter Rose operator $\mathcal{R}_{\theta,\sigma}$

Locality kernel ϕ . Let $\phi : [0, \infty) \rightarrow \mathbb{R}_+$ be radial (with respect to the Banach norm) and satisfy

$$\int_{\mathbb{R}^d} \phi(\|u\|) du = 1 \quad \phi(r) = \phi(-r) \quad \int r^2 \phi(r) dr < \infty \quad \|\phi_{\lambda_1} - \phi_{\lambda_2}\|_{L^1} \leq L |\lambda_1 - \lambda_2| \quad (13)$$

with $\phi_\lambda(r) := \lambda^{-d} \phi(r/\lambda)$. Typical choices include Gaussian, Epanechnikov, tri-weight, or a heat kernel on a Riemannian manifold; see, e.g. [30, 34, 4]. For bandwidth $\theta > 0$ define the neighbourhood weight

$$w_\theta(\|x_t - x\|) = \phi_\theta(\|x_t - x\|) \quad (14)$$

Local drift and residuals. With $W_\theta = \text{diag}(w_\theta(\|x_t - x\|))$ and $X = [x_0, \dots, x_{N-1}]$, $Y = [x_h, \dots, x_{N-1+h}]$, set

$$C_x^{(\theta)} = Y_\theta X_\theta^\top (X_\theta X_\theta^\top)^\dagger, \quad r_t^{(\theta)}(x) = x_{t+h} - C_x^{(\theta)} x_t \quad (15)$$

Residual kernel g_σ Let $g_\sigma : \mathbb{R}^d \rightarrow \mathbb{R}_+$ be radial with $\int g_\sigma = 1$ and $g_\sigma \rightarrow \delta$ as $\sigma \downarrow 0$ (e.g. Gaussian, Huber, Tukey) [30]. Define first

$$Z_{\theta,\sigma}(x) = \sum_t w_\theta(\|x_t - x\|) g_\sigma(r_t^{(\theta)}(x)) \quad (16)$$

and then the weighted residual mean

$$\bar{r}_x^{(\theta,\sigma)} = \frac{1}{Z_{\theta,\sigma}(x)} \sum_t w_\theta(\|x_t - x\|) g_\sigma(r_t^{(\theta)}(x)) r_t^{(\theta)} \quad (17)$$

as well as the covariance

$$\Sigma_x^{(\theta,\sigma)} = \frac{1}{Z_{\theta,\sigma}(x)} \sum_t w_\theta(\|x_t - x\|) g_\sigma(r_t^{(\theta)}(x)) (r_t^{(\theta)} - \bar{r}_x^{(\theta,\sigma)})(r_t^{(\theta)} - \bar{r}_x^{(\theta,\sigma)})^\top \quad (18)$$

Transition kernel.

$$k_{\theta,\sigma}(y | x) = \mathcal{N}(y; C_x^{(\theta)} x + \bar{r}_x^{(\theta,\sigma)}, \Sigma_x^{(\theta,\sigma)}) \quad (19)$$

Composition and transfer adjunctions.

Definition 3.1 (Push-forward and pull-back Rose operators). For the kernel $k_{\theta,\sigma}$ defined above set,

$$(\mathcal{R}_{\theta,\sigma}^\# \rho)(y) := \int_X k_{\theta,\sigma}(y | x) \rho(x) d\mu(x), \quad (\mathcal{R}_{\theta,\sigma}^\flat f)(x) := \int_X f(y) k_{\theta,\sigma}(y | x) d\mu(y) \quad (20)$$

Then $\mathcal{R}_{\theta,\sigma}^\flat$ is the Hilbert-space adjoint of $\mathcal{R}_{\theta,\sigma}^\#$ with respect to the $L^2(X, \mu)$ pairing,

$$\langle f, \mathcal{R}_{\theta,\sigma}^\# \rho \rangle_{L^2} = \langle \mathcal{R}_{\theta,\sigma}^\flat f, \rho \rangle_{L^2} \quad \text{for all } f, \rho \in L^2(X, \mu) \quad (21)$$

3.2 Homotopy theorems in a Banach setting

Results here need only Banach calculus (Fréchet differentiability) and the standing measure μ . Extra geometric structure (Riemannian metric, curvature proxies, and Gauss–Codazzi compatibility) will first appear in §??.

The results below make precise the idea that the two-parameter Rose operators

$$(\theta, \sigma) \mapsto (\mathcal{R}_{\theta, \sigma}^b, \mathcal{R}_{\theta, \sigma}^\#) \quad (22)$$

form a *continuous square* whose left-hand edge collapses to the Koopman / deterministic–PF pair and whose right-hand edge collapses to the rank-one, maximally diffusive PF pair.

Assumption 3.1 (Standing hypotheses).

1. X is the separable Banach space introduced earlier, endowed with finite Borel measure μ .
2. $T_h : X \rightarrow X$ is the time- h map of a C^2 (Fréchet) flow whose first and second derivatives are bounded: $\sup_x \|DT_h(x)\|_{\mathcal{L}(X)} \leq M_1$, $\sup_x \|D^2T_h(x)\| \leq M_2$.
3. The locality profile ϕ is radial, integrable, symmetric, finite-moment, and Lipschitz in scale (Section 3.1).
4. The residual-weight profile g_σ is radial with $\int g_\sigma = 1$ and $g_\sigma \rightarrow \delta$ in L^1 as $\sigma \downarrow 0$.

Theorem 3.1 (Two-parameter Rose homotopy). *Under Assumption 3.1, in the population (infinite-data) setting, the following statements hold.*

(i) **Local bias & covariance bounds**

For every $x \in X$ and $\sigma > 0$

$$\|C_x^{(\theta)} - DT_h(x)\| \leq c_1 \theta, \quad \|\Sigma_x^{(\theta, \sigma)}\| \leq c_2 (\theta^2 + \sigma^2)^2 \quad (23)$$

(ii) **Deterministic corner**

$$\lim_{\theta \rightarrow 0, \sigma \rightarrow 0} \|\mathcal{R}_{\theta, \sigma}^b - K\|_{L^2 \rightarrow L^2} = 0, \quad \lim_{\theta \rightarrow 0, \sigma \rightarrow 0} \|\mathcal{R}_{\theta, \sigma}^\# - P_\delta\|_{L^2 \rightarrow L^2} = 0 \quad (24)$$

(iii) **Maximally diffusive edge**

$$\lim_{\theta, \sigma \rightarrow \infty} \|\mathcal{R}_{\theta, \sigma}^\# - P_\infty\|_{L^2 \rightarrow L^2} = 0, \quad P_\infty \rho = \left(\int_X \rho d\mu \right) p_\infty \quad (25)$$

where p_∞ is the state-independent limit density.

(iv) **Separate Lipschitz continuity**

There exist constants $C_\theta, C_\sigma > 0$ such that

$$\|\mathcal{R}_{\theta_1, \sigma}^b - \mathcal{R}_{\theta_2, \sigma}^b\|_{L^2 \rightarrow L^2} \leq C_\theta |\theta_1 - \theta_2|, \quad \|\mathcal{R}_{\theta, \sigma_1}^b - \mathcal{R}_{\theta, \sigma_2}^b\|_{L^2 \rightarrow L^2} \leq C_\sigma \|g_{\sigma_1} - g_{\sigma_2}\|_{L^1} \quad (26)$$

and the same bounds hold for the push-forward operators $\mathcal{R}^\#$.

186 (v) **Commutativity (2-cell)**

For any $\theta_2 \geq \theta_1 > 0$ and $\sigma_2, \sigma_1 > 0$,

$$\mathcal{R}_{\theta_2, \sigma_2} = \mathcal{R}_{\theta_2, \sigma_2 - \sigma_1} \circ \mathcal{R}_{\theta_1, \sigma_1} = \mathcal{R}_{\theta_1, \sigma_1} \circ \mathcal{R}_{\theta_2 - \theta_1, \sigma_2} \quad (27)$$

188 whenever g_σ is Gaussian; more generally—by Functorial arguments in 2-category theory [22,
17, 15]—the two one-parameter homotopies commute up to a vanishing L^2 -operator error of
190 order $O(|\theta_2 - \theta_1| \|g_{\sigma_2} - g_{\sigma_1}\|_{L^1})$.

192 Constants $c_1, c_2, C_\theta, C_\sigma$ depend only on M_1, M_2 , the second moment of ϕ , and integrability bounds for g .

194 *Sketch of proof.* (i) A second-order Taylor expansion plus symmetry of ϕ yields the bias bound; the covariance bound follows by inserting $g_\sigma \leq \|g_\sigma\|_\infty$ and taking second moments.

196 (ii) As $\theta, \sigma \downarrow 0$, $C_x^{(\theta)} \rightarrow DT_h(x)$ and $\Sigma_x^{(\theta, \sigma)} \rightarrow 0$; the Gaussian TV estimate gives $k_{\theta, \sigma} \rightarrow \delta_{T_h x}$ in L^1 , hence operator convergence.

198 (iii) With $\theta, \sigma \uparrow \infty$, locality weights flatten and g_σ acts as heavy blur: the kernel columns converge to the state-independent density p_∞ .

200 (iv) Lipschitz in θ is inherited from Lipschitz scale dependence of ϕ ; Lipschitz in σ follows from $\|g_{\sigma_1} - g_{\sigma_2}\|_{L^1}$.

202 (v) For Gaussian g_σ , convolution is additive in variances, so ordering of the two dials commutes exactly; for general radial kernels one obtains the stated error estimate by Young's inequality. $\square$

$$\begin{array}{ccc} (K, P_\delta) & \xrightarrow{\sigma} & (R_{\theta, \infty}^b, R_{\theta, \infty}^\#) \\ \theta \downarrow & \boxed{H(\theta, \sigma)} & \downarrow \theta \\ (R_{\infty, \sigma}^b, R_{\infty, \sigma}^\#) & \xrightarrow{\sigma} & (P_\infty^*, P_\infty) \end{array}$$

Figure 1: Homotopy square for 2-cell (homotopy of homotopies) that continuously deforms the Koopman operator K into the maximally diffusive transfer operator P_∞ .

4 Continuous–Time Limit and Stochastic Interpretation

204 To speak of gradients, divergences, and *Itô diffusions* we now fix $X = \mathbb{R}^d$ equipped with its Euclidean norm $\|\cdot\|_2$ and the same reference measure μ [16, 25]. All Fréchet derivatives therefore
206 coincide with classical derivatives; no Riemannian structure is yet required.

Apart from this dimensionality pin-down, all Banach-level hypotheses from Assumption 3.1 remain in force.

Generality of the limit. Throughout this section we work *solely* with norms and local linearizations; no Riemannian structure is required. All results therefore hold in any Banach state space. In §?? we specialize to spaces that *do* admit a (possibly synthetic) Riemannian metric so that curvature can be matched to residual diffusion.

The discrete-time Rose operators $(\mathcal{R}_{\theta,\sigma}^b, \mathcal{R}_{\theta,\sigma}^\sharp)$ approximate, in the small-lag limit, the semigroup of an Itô diffusion [7]. This section formalises that link and shows how the local bandwidth-weighted quantities $C_x^{(\theta)}$ and $\Sigma_x^{(\theta,\sigma)}$ furnish non-parametric estimators of the drift $a(x)$ and diffusion tensor $D(x)$ of the underlying continuous dynamics.

Throughout, $h > 0$ is the sampling lag. We write $C_x^{(\theta,h)}$, $r_t^{(\theta,h)}$, $\Sigma_x^{(\theta,\sigma,h)}$ to expose this dependence explicitly.

Assumptions 3.1 remain in force.

Lemma 4.1 (First-order (small- h) expansion). *For every fixed $\theta > 0, \sigma > 0$ and anchor $x \in \mathbb{R}^d$*

$$C_x^{(\theta,h)} = I + a(x)h + O(h\theta + h^2), \quad \Sigma_x^{(\theta,\sigma,h)} = 2D(x)h + O(h\theta^2 + h\sigma^2) \quad (28)$$

where $a(x) = \dot{X}_t|_{t=0}$ is the deterministic drift and $D(x) = \frac{1}{2}B(x)B(x)^\top$ is the diffusion tensor of the Itô SDE $dX_t = a(X_t)dt + B(X_t)dW_t$ [11, 14].

Sketch. Second-order Taylor expansion of T_h yields $T_h x_t = x + a(x)h + O(h\|x_t - x\| + h^2)$. Weighted first-moment symmetry of ϕ eliminates the linear $x_t - x$ term, giving the stated drift expansion. For the covariance, subtract $\bar{r}_x^{(\theta,\sigma,h)}$ and take second moments; the leading term is $2D(x)h$ and the weights introduce the stated $O(h\theta^2 + h\sigma^2)$ correction. $\square$

Theorem 4.1 (Discrete-to-continuous limit). *Fix $\theta, \sigma > 0$ and let $h \downarrow 0$. For any $f \in C^2(X)$ and $\rho \in C^2(X)$,*

$$\frac{\mathcal{R}_{\theta,\sigma}^{b,h} f - f}{h} \longrightarrow \mathcal{L}^b f, \quad \frac{\mathcal{R}_{\theta,\sigma}^{\sharp,h} \rho - \rho}{h} \longrightarrow \mathcal{L}^\sharp \rho \quad \text{in } L^2(X, \mu) \quad (29)$$

where the Kolmogorov generators are

$$\mathcal{L}^b f = a \cdot \nabla f + D : \nabla \nabla f, \quad \mathcal{L}^\sharp \rho = -\nabla \cdot (a\rho) + \nabla \cdot (D \nabla \rho) \quad (30)$$

corresponding to the Itô SDE above.

Sketch. Expand $f(y)$ to second order around x and integrate against the Gaussian kernel with mean $x + a(x)h$ and variance $2D(x)h$; the first-order terms reproduce $\mathcal{L}^b f$. An identical calculation with adjoint pairing yields the forward generator $\mathcal{L}^\sharp$. $\square$

Remark 4.1 (Non-parametric drift/diffusion estimators). Lemma 4.1 yields *data-driven, bandwidth-controlled* estimators for both coefficients of the limiting SDE, namely $\hat{a}(x) = (C_x^{(\theta,h)} - I)/h$ for the drift and $\hat{D}(x) = \frac{1}{2} \Sigma_x^{(\theta,\sigma,h)}/h$ for the diffusion tensor. Once these fields are available, the generator pair

($\mathcal{L}^b, \mathcal{L}^\#$) provides two complementary routes to uncertainty propagation: forward evolution of densities through the Fokker–Planck equation and simulation of sample paths via the Itô SDE, with the discrete Rose operators furnishing time- h stepping schemes that are consistent with both perspectives.

5 Metric–measure enrichment: from Banach to curvature

The results in §§1–4 required only that $(X, \|\cdot\|_X)$ be a separable Banach space equipped with a finite reference measure μ . We now introduce additional geometric structure. Everything in §3.2 remains valid; the extra structure is used to motivate and constrain the (θ, σ) selection.

So far the two dials (θ, σ) have been tuned independently. When residual variance is truly homoscedastic that is appropriate, but if the unexplained noise depends on state we can—and should—insist on a *compatibility* between the *metric* furnished by the drift weights and the *second fundamental form* furnished by the diffusion covariance, echoing the classical Gauss–Codazzi equations for an embedded C^2 manifold [5, 31].

5.1 Complete, separable metric–measure space

Assumption 5.1 (Metric–measure setting).

- (i) (X, d) is a complete, separable metric space.
- (ii) μ is a finite Borel measure with full support.

Under Assumption 5.1, heat-kernel and optimal-transport tools provide *synthetic curvature* quantities that reduce to classical sectional or Ricci curvature whenever a smooth metric exists; see, e.g. the Curvature–Dimension $\text{CD}(K, N)$ condition of Lott–Sturm–Villani [32, 21] or the Ollivier–Ricci curvature [26] κ_{OR} . We fix once and for all any curvature functional

$$\mathsf{K}_{\text{synthetic}} : \{\text{prob. kernels on } X\} \longrightarrow \mathbb{R}, \quad (31)$$

such that $\mathsf{K}_{\text{synthetic}}(\delta_x) = 0$ and $\mathsf{K}_{\text{synthetic}}$ is Lipschitz with respect to the L^1 distance between kernels.

Metric–measure curvature proxy. Given a localization bandwidth θ at anchor x we regard the weighted Laplacian generated by the kernel w_θ as a discrete heat operator and define

$$K_{\text{mm}}(\theta; x) := \mathsf{K}_{\text{synthetic}}(w_\theta(\cdot \mid x)). \quad (32)$$

This quantity is well-defined in every complete, separable metric–measure space and converges to the usual Ricci trace when a C^2 Riemannian metric is present.

Residual isotropy remains purely algebraic: $K_{\text{noise}}(\sigma; x)$ is still the Frobenius-norm defect of the empirical covariance (§??).

266 **Synthetic Gauss–Codazzi loss.** With anchors $\{x^{(m)}\}_{m=1}^M$ we keep the same penalty

$$E_{\text{GC}}^{\text{synthetic}} = \sum_{m=1}^M \left(K_{\text{mm}}(\theta; x^{(m)}) - K_{\text{noise}}(\sigma; x^{(m)}) \right)^2, \quad (33)$$

now interpreted as a *regularizer* rather than an exact compatibility condition—there is no ambient
268 smooth manifold yet, but we can still force the two “curvature witnesses” to agree as closely as the data allow.

270 5.2 Embedded C^2 manifold

Assume in addition that

$$X \subset \mathbb{R}^D \text{ is an embedded } C^2 \text{ sub-manifold}$$

272 with inherited Euclidean metric g . Denote by ∇ , Ric , etc. the usual geometric objects.

First fundamental form from localisation. For any anchor x let $\{\xi_i\}_{i=1}^d$ be an orthonormal
274 basis of $T_x X$. Define the *local metric tensor*

$$g_{ij}^{(\theta)}(x) := \frac{1}{Z_\theta(x)} \sum_t w_\theta(\|x_t - x\|) \langle x_t - x, \xi_i \rangle \langle x_t - x, \xi_j \rangle, \quad Z_\theta(x) = \sum_t w_\theta(\|x_t - x\|).$$

Up to $O(\theta^2)$ this is the periodised heat kernel estimate of $\exp(-\theta^2 \Delta)g$; hence $g^{(\theta)} \rightarrow g$ as $\theta \downarrow$
276 0.

Second fundamental form from residuals. With the same notation,

$$\Pi_{ij}^{(\sigma)}(x) := \frac{1}{Z_{\theta,\sigma}(x)} \sum_t w_\theta(\|x_t - x\|) g_\sigma(r_t^{(\theta)}) \langle r_t^{(\theta)}, \xi_i \rangle \langle r_t^{(\theta)}, \xi_j \rangle.$$

278 When the unexplained dynamics are pure measurement noise and the drift map is bias-free, $\Pi^{(\sigma)}$ is (up to isotropic scaling) the extrinsic second fundamental form of the data-manifold embed-
280 ding.

Discrete Gauss–Codazzi discrepancy. Write $\text{Ric}^{(\theta)}$ for the Ricci tensor of $g^{(\theta)}$ and $\text{tr}_{g^{(\theta)}} \Pi^{(\sigma)}$
282 for the mean-curvature contraction. The classical *Gauss–Codazzi* relation is $\text{Ric} - \Pi \equiv 0$ in codimension one. We therefore refine §?? by setting

$$E_{\text{GC}}^{\text{manifold}} := \sum_{m=1}^M \left\| \text{Ric}^{(\theta)}(x^{(m)}) - \text{tr}_{g^{(\theta)}} \Pi^{(\sigma)}(x^{(m)}) \right\|_F^2,$$

284 and continue to optimise $J(\theta, \sigma) = L_{\text{drift}} + L_{\text{diff}} + \lambda E_{\text{GC}}^{\text{manifold}}$.

Remark. Because $g^{(\theta)}$ and $\Pi^{(\sigma)}$ converge (in the infinite-sample, $\theta, \sigma \rightarrow 0$ limit) to their smooth
286 counterparts, the discrete mismatch $E_{\text{GC}}^{\text{manifold}}$ vanishes at first order when the model is noise-free and the manifold assumption is correct; otherwise it quantifies the *net curvature still hiding in the
288 residuals*.

What comes next?

Sections §??–?? can now be read in two flavours:

* under Assumption 5.1 with $E_{GC}^{\text{synthetic}}$ (regulariser); * under the manifold upgrade with E_{GC}^{manifold} (compatibility constraint).

Further geometric enrichments (e.g. non-smooth Alexandrov spaces, simplicial complexes, or sub-Riemannian structures) fit by swapping in the corresponding synthetic curvature in $K_{\text{synthetic}}$.

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360 A Notation & Conventions

PX	probability measures on $(X, \mathcal{F})$
G	Giry monad (B.15)
$x \mapsto \delta_x$	Dirac embedding
$\rightsquigarrow$	kernel arrow (Def. B.13)
$\flat, \sharp$	musical conversions (Def. ??)
P_t, L	Markov semigroup & generator (B.23–B.24)
k_Δ, T	LWLS kernel & Rose operator (??–??)
Δ	discrete time step in data

362 B Definitions

B.1 Measure & Probability Foundations

364 **Definition B.1** (σ -Algebra). Given a set X , a σ -algebra $\mathcal{F} \subseteq 2^X$ is a non-empty collection of subsets closed under complementation and countable unions (hence also countable intersections).

366 **Definition B.2** (Measurable Space). A *measurable space* is a pair $(X, \mathcal{F})$ where $\mathcal{F}$ is a σ -algebra on X (Definition B.1).

368 **Definition B.3** (Measurable Map). If $(X, \mathcal{F})$ and $(Y, \mathcal{G})$ are measurable spaces, a map $f : X \rightarrow Y$ is *measurable* if $f^{-1}(B) \in \mathcal{F}$ for all $B \in \mathcal{G}$.

370 **Definition B.4** (Measure & Probability Measure). Let $(X, \mathcal{F})$ be a measurable space. A *measure* is a countably additive map $\mu : \mathcal{F} \rightarrow [0, \infty]$ with $\mu(\emptyset) = 0$. It is a *probability measure* if $\mu(X) = 1$.
372 We write PX for the set of all probability measures on $(X, \mathcal{F})$.

374 **Definition B.5** (Push-Forward Measure). Given a measurable $f : (X, \mathcal{F}) \rightarrow (Y, \mathcal{G})$ and $\mu \in PX$, the *push-forward* measure $f_*\mu \in PY$ is defined by $(f_*\mu)(B) = \mu(f^{-1}(B))$ for $B \in \mathcal{G}$.

376 **Definition B.6** (Pull-Back Measure). Given a measurable $f : (X, \mathcal{F}) \rightarrow (Y, \mathcal{G})$ and $\mu \in PY$, the *pull-back* measure $f^*\mu \in PX$ is defined by $f^*\mu(A) := \mu(f(A))$ for $A \in \mathcal{F}$.

B.2 Category Theory

378 **Definition B.7** (Category). A *category* $\mathcal{C}$ consists of a class of objects, hom-sets $\text{Hom}_{\mathcal{C}}(A, B)$ for each ordered pair (A, B) , identity morphisms id_A , and associative composition, satisfying the usual
380 unit laws.

382 **Definition B.8** (Functor). Let $\mathcal{C}, \mathcal{D}$ be categories. A *functor* $F : \mathcal{C} \rightarrow \mathcal{D}$ assigns to each object A of $\mathcal{C}$ an object FA of $\mathcal{D}$ and to each morphism $f : A \rightarrow B$ a morphism $Ff : FA \rightarrow FB$, preserving identities and composition.

384 **Definition B.9** (Finite products in a category). Let $\mathcal{C}$ be a category and $\{X_i\}_{i=1}^n$ a finite family of objects in $\mathcal{C}$ ($n \geq 0$).

- A *product* of the family is a pair

$$\left(X, \{ \pi_i \}_{i=1}^n \right), \quad \pi_i : X \longrightarrow X_i \text{ in } \mathcal{C},$$

such that for every object Y and every collection of morphisms $\{ f_i : Y \rightarrow X_i \}_{i=1}^n$ there exists a *unique* morphism

$$f : Y \longrightarrow X \quad \text{with} \quad \pi_i \circ f = f_i \quad (i = 1, \dots, n).$$

- When it exists, the product is unique up to a unique isomorphism; we write $X = \prod_{i=1}^n X_i$ and call the π_i the *projection morphisms*.
- For $n = 0$ the empty product is any *terminal object* $1 \in \mathcal{C}$ (unique up to unique isomorphism).

$$\begin{array}{ccccc} & & Y & & \\ & f_1 \swarrow & \downarrow f & \searrow f_2 & \\ X_1 & \xleftarrow{\pi_1} & X_1 \times X_2 & \xrightarrow{\pi_2} & X_2 \end{array}$$

The dashed arrow f is the *unique* morphism rendering all triangles commutative: $\pi_i \circ f = f_i$ for $i = 1, \dots, n$. This diagram visualises the universal property that characterises the product object $X = \prod_{i=1}^n X_i$.

Definition B.10 (Natural Transformation). Given functors $F, G : \mathcal{C} \rightarrow \mathcal{D}$, a *natural transformation* $\eta : F \Rightarrow G$ assigns to every object A a morphism $\eta_A : FA \rightarrow GA$ such that $Gf \circ \eta_A = \eta_B \circ Ff$ for all $f : A \rightarrow B$ in $\mathcal{C}$.

Definition B.11 (Monad). A *monad* (T, η, μ) on a category $\mathcal{C}$ comprises an endofunctor $T : \mathcal{C} \rightarrow \mathcal{C}$, a unit natural transformation $\eta : \text{id}_{\mathcal{C}} \Rightarrow T$ and a multiplication $\mu : T^2 \Rightarrow T$ satisfying $\mu \circ T\eta = \mu \circ \eta T = \text{id}_T$ and $\mu \circ T\mu = \mu \circ \mu T$.

Definition B.12 (Category **Meas**). Objects are measurable spaces; morphisms are measurable maps (Definition B.3).

Definition B.13 (Markov Kernel). Let $(X, \mathcal{F})$ and $(Y, \mathcal{G})$ be measurable spaces. A *Markov kernel* $\kappa : X \rightsquigarrow Y$ is a map $\kappa : X \times \mathcal{G} \rightarrow [0, 1]$ such that

1. for each $x \in X$, $B \mapsto \kappa(B \mid x)$ is a probability measure on $(Y, \mathcal{G})$;
2. for each $B \in \mathcal{G}$, $x \mapsto \kappa(B \mid x)$ is $\mathcal{F}$ -measurable.

Definition B.14 (Category **Stoch**). Objects are measurable spaces. Morphisms are Markov kernels (Definition B.13). Composition is *Chapman–Kolmogorov* integration:

$$(\kappa * \lambda)(C \mid x) = \int_Y \kappa(C \mid y) \lambda(dy \mid x).$$

Definition B.15 (Giry Monad). Define $G : \mathbf{Meas} \rightarrow \mathbf{Meas}$ by $G(X, \mathcal{F}) = (PX, \mathcal{PF})$, where $\mathcal{PF}$ is the weak σ -algebra generated by $\mu \mapsto \mu(B)$ ($B \in \mathcal{F}$). For $f : (X, \mathcal{F}) \rightarrow (Y, \mathcal{G})$ set $Gf = f_*$

(Definition B.5). The *unit* $\eta_X : X \rightarrow GX$ is $x \mapsto \delta_x$ and the *multiplication* $\mu_X : GGX \rightarrow GX$ is $\mu_X(\Phi)(B) = \int_{PX} \nu(B) d\Phi(\nu)$. With these, (G, η, μ) is a monad on **Meas** (Definition B.11); its Kleisli category is **Stoch** (Definition B.14).

Definition B.16 (Stochastic Dynamical System). A *stochastic dynamical system* is a G -coalgebra: a measurable map $c : X \rightarrow GX$ (Definition B.15). Iterating c via Kleisli composition yields a family of kernels $c^{(n)} : X \rightsquigarrow X$; see Definition B.17.

B.3 Markov Kernels & Stochastic Processes

Definition B.17 (Transition Kernel Family). A family $\{\kappa_t\}_{t \geq 0}$ of kernels $\kappa_t : X \rightsquigarrow X$ is a *transition kernel family* if

$$\kappa_{t+s} = \kappa_t * \kappa_s, \quad \kappa_0(x, \cdot) = \delta_x.$$

Definition B.18 (Discrete-Time Markov Chain). A *discrete-time Markov chain* on state space X is a pair (κ, μ_0) where $\kappa : X \rightsquigarrow X$ is a kernel and $\mu_0 \in PX$ an initial distribution. The law of X_{n+1} given $X_n = x$ is $\kappa(\cdot | x)$.

B.4 Semigroup & Generator Theory

Definition B.19 (Bounded Linear Operators). For a Banach space B , the set $\mathcal{L}(B)$ consists of all linear maps $A : B \rightarrow B$ with finite operator norm $\|A\|_{\mathcal{L}(B)} := \sup_{f \neq 0} \frac{\|Af\|_B}{\|f\|_B}$.

Equipped with this norm, is itself a Banach space.

Definition B.20 (Operator Semigroup). Let B be a Banach space. A family $\{P_t\}_{t \geq 0} \subset \mathcal{L}(B)$ is called an *operator semigroup* (on B) if

$$P_0 = \text{id}_B, \quad P_{t+s} = P_t P_s \quad \text{for all } s, t \geq 0.$$

Definition B.21 (Uniformly Continuous Semigroup). A family $\{P_t\}_{t \geq 0} \subset \mathcal{L}(B)$ is a *uniformly continuous semigroup* if

$$\lim_{t \downarrow 0} \|P_t - \text{id}\|_{\mathcal{L}(B)} = 0, \quad P_0 = \text{id}, \quad \text{for all } t \geq 0$$

Equivalently, $P_t = e^{tA}$ for a *bounded* operator $A \in \mathcal{L}(B)$ called the (bounded) generator, and the map $t \mapsto P_t$ is continuous in operator norm.

Definition B.22 (Strongly Continuous Semigroup). A family $\{P_t\}_{t \geq 0} \subset \mathcal{L}(B)$ is a *strongly continuous semigroup* (C_0 -semigroup) if

$$\lim_{t \downarrow 0} \|P_t f - f\|_B = 0, \quad \text{for all } f \in B, t \geq 0$$

Definition B.23 (Markov Semigroup). A C_0 -semigroup $\{P_t\}_{t \geq 0}$ on $L^p(X, \lambda)$ is a *Markov semigroup* if each P_t is positive and $P_t 1 = 1$.

Definition B.24 (Infinitesimal Generator). For a C_0 -semigroup $\{P_t\}_{t \geq 0}$ the *generator* is the (possibly unbounded) operator

$$Lf = \lim_{t \downarrow 0} \frac{P_t f - f}{t},$$

with domain $D(L) = \{f : \text{limit exists in } \|\cdot\|\}$.

C Locality in Infinitesimal Generators

C.1 The Evolution Kernel

In the classical theory of C_0 semigroups, the evolution operator $T(h)$ composes a C_0 semigroup generated by a linear operator A

$$T(h) = e^{hA}.$$

In this setting, the kernel $K_h(\xi, \eta)$ is the Dirac delta

$$K_h(\xi, \eta) \xrightarrow{h \rightarrow 0} \delta(\xi - \eta).$$

Proof. Consider the integral operator

$$(T(h)f)(\xi) = \int_{\Omega} K_h(\xi, \eta) f(\eta) d\eta \quad (34)$$

with infinitesimal generator

$$(Af)(\xi) = \lim_{h \rightarrow 0} \frac{1}{h} (T(h) - I) f(\xi) \quad (35)$$

$$= \lim_{h \rightarrow 0} \frac{1}{h} ((T(h)f)(\xi) - f(\xi)) \quad (36)$$

By substitution

$$(Af)(\xi) = \lim_{h \rightarrow 0} \frac{1}{h} \left(\int_{\Omega} K_h(\xi, \eta) f(\eta) d\eta - f(\xi) \right) \quad (37)$$

The goal is to rewrite the integral term so that $f(\xi)$ appears inside the integral. We add and subtract $K_h(\xi, \eta) f(\xi)$ inside the integral,

$$(Af)(\xi) = \lim_{h \rightarrow 0} \frac{1}{h} \left(\int_{\Omega} K_h(\xi, \eta) f(\eta) - K_h(\xi, \eta) f(\xi) + K_h(\xi, \eta) f(\xi) d\eta - f(\xi) \right) \quad (38)$$

and factor,

$$(Af)(\xi) = \lim_{h \rightarrow 0} \frac{1}{h} \left(\int_{\Omega} K_h(\xi, \eta) (f(\eta) - f(\xi)) d\eta + \int_{\Omega} K_h(\xi, \eta) f(\xi) d\eta - f(\xi) \right) \quad (39)$$

$$= \lim_{h \rightarrow 0} \frac{1}{h} \int_{\Omega} K_h(\xi, \eta) (f(\eta) - f(\xi)) d\eta + \lim_{h \rightarrow 0} \frac{1}{h} \left(\int_{\Omega} K_h(\xi, \eta) d\eta - 1 \right) f(\xi) \quad (40)$$

We can ensure that $K_h(\xi, \eta)$ acts as a state transition probability density by imposing the normalization condition,

$$\int_{\Omega} K_h(\xi, \eta) d\eta = 1 \quad (41)$$

while at the same time eliminating the second term,

$$\lim_{h \rightarrow 0} \frac{1}{h} \left(\int_{\Omega} K_h(\xi, \eta) d\eta - 1 \right) f(\xi) = 0 \quad (42)$$

thus recovering the infinitesimal generator as a standard non-local operator,

$$(Af)(\xi) = \lim_{h \rightarrow 0} \frac{1}{h} \int_{\Omega} K_h(\xi, \eta) (f(\eta) - f(\xi)) d\eta \quad (43)$$

Kernel Structure: The resolvent operator associated with A is given by:

$$R_{\lambda}(A) = (\lambda I - A)^{-1}. \quad (44)$$

It is well known (see, e.g. Pazy) that the semigroup $T(h)$ can be expressed in terms of the resolvent via the contour integral,

$$T(h) = \frac{1}{2\pi i} \oint_{\Gamma} e^{h\lambda} R_{\lambda}(A) d\lambda \quad (45)$$

We equate the integral representations

$$\int_{\Omega} K_h(\xi, \eta) f(\eta) d\eta = \frac{1}{2\pi i} \oint_{\Gamma} e^{h\lambda} R_{\lambda}(A) f(\xi) d\lambda \quad (46)$$

Furthermore, Arendt & Bukhvalov showed that if $T(h)$ is an integral operator, then so is the resolvent,

$$(R_{\lambda}f)(\xi) = \int_{\Omega} M_{\lambda}(\xi, \eta) f(\eta) d\eta \quad (47)$$

where $M_{\lambda}(\xi, \eta)$ is the resolvent kernel. From (46) using the integral resolvent,

$$\int_{\Omega} K_h(\xi, \eta) f(\eta) d\eta = \frac{1}{2\pi i} \oint_{\Gamma} e^{h\lambda} \int_{\Omega} M_{\lambda}(\xi, \eta) f(\eta) d\eta d\lambda \quad (48)$$

Two cases follow regarding the absolute integrability of

$$F(\lambda, \eta) = e^{h\lambda} M_{\lambda}(\xi, \eta) f(\eta)$$

If $F \in L^1(\Gamma \times \Omega)$ then for

$$\oint_{\Gamma} \int_{\Omega} |F(\lambda, \eta)| d\mu(\eta) d\nu(\lambda) \quad (49)$$

458 we may apply Fubini's theorem and interchange the order of integration

$$\int_{\Omega} K_h(\xi, \eta) f(\eta) d\eta = \int_{\Omega} \left(\frac{1}{2\pi i} \oint_{\Gamma} e^{h\lambda} M_{\lambda}(\xi, \eta) d\lambda \right) f(\eta) d\eta \quad (50)$$

Since this must hold for all f , we identify the kernels,

$$K_h(\xi, \eta) = \frac{1}{2\pi i} \oint_{\Gamma} e^{h\lambda} M_{\lambda}(\xi, \eta) d\lambda \quad (51)$$

460 Since $T(0) = I$, the kernel at $h = 0$ must satisfy

$$K_0(\xi, \eta) = \delta(\xi - \eta) \quad (52)$$

In many classical settings, one can show that the resolvent satisfies

$$(\lambda I - A)R_{\lambda}(A) = I \quad (53)$$

462 so that in kernel form

$$(\lambda I - A)M_{\lambda}(\xi, \eta) = \delta(\xi - \eta) \quad (54)$$

$M_{\lambda}(\xi, \eta)$ serves as the Green's function for the operator $\lambda I - A$. Supposing that for values of λ in a suitable region the Neumann series expansion hold

$$(\lambda I - A)^{-1} = \sum_{n=0}^{\infty} \frac{A^n}{\lambda^{n+1}} \quad (55)$$

464 Then the kernel $M_{\lambda}(\xi, \eta)$ may be expressed as

$$(56)$$

$$M_{\lambda}(\xi, \eta) = \sum_{n=0}^{\infty} \frac{A^n}{\lambda^{n+1}} \delta(\xi - \eta) \quad (57)$$

Substitute this expansion into

$$K_h(\xi, \eta) = \frac{1}{2\pi i} \oint_{\Gamma} e^{h\lambda} \left(\sum_{n=0}^{\infty} \frac{A^n}{\lambda^{n+1}} \delta(\xi - \eta) \right) d\nu(\lambda) \quad (58)$$

$$= \delta(\xi - \eta) \sum_{n=0}^{\infty} A^n \left[\frac{1}{2\pi i} \oint_{\Gamma} \frac{e^{h\lambda}}{\lambda^{n+1}} d\nu(\lambda) \right] \quad (59)$$

466 According to the residue theorem

$$\frac{1}{2\pi i} \oint_{\Gamma} \frac{e^{h\lambda}}{\lambda^{n+1}} d\nu(\lambda) = \frac{h^n}{n!} \quad (60)$$

Substituting this back into (59), we obtain

$$K_h(\xi, \eta) = \delta(\xi - \eta) \sum_{n=0}^{\infty} \frac{h^n}{n!} A^n = \delta(\xi - \eta) e^{hA} \quad (61)$$

468 Thus,

$$K_h(\xi, \eta) = \delta(\xi - \eta)e^{hA} \quad (62)$$

which is equivalent to the operator identity,

$$T(h) = e^{hA} \quad (63)$$

470

□

472 **The Nonlinear (State-Dependent) Case:** When the system is nonlinear, the evolution is typically given by a state-dependent mapping,

$$T(h)(\xi) = \Phi_h(\xi),$$

we may still maintain an integral representation:

$$(T(h)f)(\xi) = \int_{\Omega} K_h(\xi, \eta) f(\eta) d\eta.$$

474 In order for the kernel to capture the correct local behavior of the flow, we require that for small h ,

$$K_h(\xi, \eta) \rightarrow \delta(\Phi_h(\xi) - \Phi_h(\eta)).$$

476 Since Φ_0 is the identity mapping, this condition reduces to

$$\lim_{h \rightarrow 0} K_h(\xi, \eta) = \delta(\xi - \eta),$$

which, in the weak sense, means that for any test function f ,

$$\lim_{h \rightarrow 0} \int_{\Omega} K_h(\xi, \eta) f(\eta) d\eta = f(\xi).$$

478 D A Categorical Scaffold for Local Linearization

Local linear models, residual covariances, and adaptive basis updates can all be written algebraically, but a categorical formulation clarifies *which parts belong to the dynamics, which to the coordinate representation, and which to the data-driven closure*. By lifting the construction to functors, natural transformations, and (co)monads we obtain

- (i) a precise separation of concerns
- 484 (ii) a route to operator-theoretic results via functoriality
- (iii) and a universal property (*cofree* coalgebra) that characterises the residual-frame augmentation

486 D.1 Components

Throughout, $k \geq 2$ is fixed, so that C^k flows have well-defined first and second jets. All manifolds
488 are assumed compact, and all vector spaces are over $\mathbb{R}$.

Definition D.1 (Dyn). The category **Dyn** consists of

- 490 • **objects** $(\mathcal{M}, \varphi^t)$ where $\mathcal{M}$ is a compact C^k manifold and $\varphi^t : \mathcal{M} \rightarrow \mathcal{M}$ a C^k flow.
- **morphisms** $\Psi : (\mathcal{M}, \varphi^t) \rightarrow (\mathcal{N}, \psi^t)$ given by C^k maps $\Psi : \mathcal{M} \rightarrow \mathcal{N}$ satisfying

$$\Psi \circ \varphi^t = \psi^t \circ \Psi \quad (64)$$

492 for all $t \in \mathbb{R}$.

Remark D.1. Objects in **Dyn** are smooth dynamical systems. Morphisms Ψ are *dynamic-preserving*
494 (equivalently, *equivariant* or *trajectory-intertwining*). They are dynamic-preserving in the sense that for smooth manifolds and flows:

$$(\mathcal{M}, \varphi^t), (\mathcal{N}, \psi^t) \quad t \in \mathbb{R} \quad (65)$$

496 the morphism,

$$\Psi : (\mathcal{M}, \varphi^t) \rightarrow (\mathcal{N}, \psi^t) \quad (66)$$

is a C^k ($k \geq 1$) map that is equivariant,

$$\Psi(\varphi^t(x)) = \psi^t(\Psi(x)) \quad \forall x \in \mathcal{M} \quad (67)$$

498 so that Ψ transports trajectories φ^t onto trajectories ψ^t :

$$\Psi \circ \varphi^t = \psi^t \circ \Psi \quad (68)$$

Example D.1. Take the two-torus with linear flow,

$$(\mathbb{T}^2, \varphi^t), \quad \varphi^t(x, y) = (x + \alpha t, y + \beta t) \pmod{1} \quad (69)$$

500 and the circle with rigid rotation,

$$(\mathbb{S}^1, \psi^t), \quad \psi^t(\theta) = \theta + \alpha t \pmod{1} \quad (70)$$

Let,

$$\Psi : \mathbb{T}^2 \rightarrow \mathbb{S}^1, \quad \Psi(x, y) = x \pmod{1} \quad (71)$$

502 For all $(x, y) \in \mathbb{T}^2$ $t \in \mathbb{R}$:

$$\Psi(\varphi^t(x, y)) = \Psi(x + \alpha t, y + \beta t) \quad (72)$$

$$= (x + \alpha t) \pmod{1} \quad (73)$$

$$= \psi^t(\Psi(x, y)) \quad (74)$$

so,

$$\Psi \circ \varphi^t = \psi^t \circ \Psi, \quad \forall t \in \mathbb{R} \quad (75)$$

504 and Ψ is a dynamic-preserving C^∞ map and morphism of **Dyn**.

506 *Remark D.2* (Discrete-time systems via suspension). Although **Dyn** is defined only for *flows*, any discrete-time diffeomorphism $T: \mathcal{M} \rightarrow \mathcal{M}$ can be embedded in **Dyn** through its *suspension flow*.

Form the mapping torus

$$\text{Sus}(\mathcal{M}, T) := (\mathcal{M} \times [0, 1]) / (x, 1) \sim (Tx, 0), \quad (76)$$

508 and define the flow

$$\varphi^t[(x, u)] = [(T^{\lfloor t+u \rfloor} x, u + t - \lfloor t+u \rfloor)] \quad (77)$$

Then $(\text{Sus}(\mathcal{M}, T), \varphi^t)$ is an object of **Dyn** whose *time-1 map* reproduces T :

$$\varphi^1[(x, 0)] = [(Tx, 0)] \quad (78)$$

510 This construction extends to morphisms by $\Sigma(f)[(x, u)] = [(f(x), u)]$, producing a functor $\Sigma: \mathbf{Dyn}_{\mathbb{Z}} \rightarrow \mathbf{Dyn}$. As such, results for continuous-time objects apply *mutatis mutandis* to discrete systems via Σ .
512

Definition D.2 (**Vect**). The category **Vect** consists of

- 514 • **objects** V finite-dimensional real vector spaces.
- **morphisms** $L: V \rightarrow W$ linear maps.

516 **Definition D.3** (Embedding functor). Fix, for every Dyn-object $(\mathcal{M}, \varphi^t)$ a generic embedding $\Phi_{\mathcal{M}}: \mathcal{M} \hookrightarrow \mathbb{R}^{m(\mathcal{M})}$. Define the functor

$$F: \mathbf{Dyn} \longrightarrow \mathbf{Vect} \quad \text{by} \quad \begin{cases} F(\mathcal{M}, \varphi^t) = \mathbb{R}^{m(\mathcal{M})}, \\ F(\Psi) = \Phi_{\mathcal{N}} \circ \Psi \circ \Phi_{\mathcal{M}}^{-1}: \mathbb{R}^{m(\mathcal{M})} \longrightarrow \mathbb{R}^{m(\mathcal{N})}. \end{cases}$$

518 with identity $F(\text{id}_{(\mathcal{M}, \varphi^t)}) = \text{id}_{\mathbb{R}^{m(\mathcal{M})}}$.

Functoriality is immediate from $\Phi_{\mathcal{N}} \circ (\Psi_2 \circ \Psi_1) = F(\Psi_2) \circ F(\Psi_1)$, which gives the following diagram:

$$\begin{array}{ccc} (\mathcal{M}, \varphi^t) & \xrightarrow{\Phi := \Phi_{\mathcal{M}}} & \mathbb{R}^{m(\mathcal{M})} \\ \Psi \downarrow & & \downarrow \Phi' \circ \Psi \circ \Phi^{-1} \\ (\mathcal{N}, \psi^t) & \xrightarrow{\Phi' := \Phi_{\mathcal{N}}} & \mathbb{R}^{m(\mathcal{N})} \end{array}$$

520 **Definition D.4** (**Dyn** Component-wise product). Let $(\mathcal{M}_i, \varphi_i^t)_{i=1}^n$ be objects of **Dyn**. Define

$$\mathcal{M} := \prod_{i=1}^n \mathcal{M}_i, \quad \varphi^t(x_1, \dots, x_n) := (\varphi_1^t(x_1), \dots, \varphi_n^t(x_n)), \quad (79)$$

and projection maps $\pi_j(x_1, \dots, x_n) = x_j$ ($j = 1, \dots, n$). We call

$$(\mathcal{M}, \varphi^t) \quad (80)$$

522 the *component-wise product* of the given objects.

Lemma D.1 (Universal property of finite products in **Dyn**). *For the system $(\mathcal{M}, \varphi^t)$ and projections π_j defined above, the pair*

$$((\mathcal{M}, \varphi^t), \{\pi_j\}_{j=1}^n) \quad (81)$$

*is the categorical product of $(\mathcal{M}_i, \varphi_i^t)_{i=1}^n$ in the category **Dyn**.*

Proof. Because each π_j is smooth,

$$\pi_j(\varphi^t(x_1, \dots, x_n)) = \varphi_j^t(x_j) = \varphi_j^t(\pi_j(x_1, \dots, x_n)), \quad (82)$$

so π_j intertwines the flows and is therefore a morphism in **Dyn**.

Let $(\mathcal{N}, \psi^t)$ be any object and suppose we are given morphisms $f_j : (\mathcal{N}, \psi^t) \rightarrow (\mathcal{M}_j, \varphi_j^t)$ for $j = 1, \dots, n$. Define

$$f : \mathcal{N} \longrightarrow \mathcal{M}, \quad f(y) := (f_1(y), \dots, f_n(y)). \quad (83)$$

The map f is C^k because each f_j is. Equivariance follows component-wise:

$$f(\psi^t y) = (f_1(\psi^t y), \dots) = (\varphi_1^t f_1(y), \dots) = \varphi^t(f(y)). \quad (84)$$

Hence f is a morphism in **Dyn** and $\pi_j \circ f = f_j$ by construction.

If $g : (\mathcal{N}, \psi^t) \rightarrow (\mathcal{M}, \varphi^t)$ is another morphism with the same property $\pi_j \circ g = f_j$ for every j , then for each $y \in \mathcal{N}$

$$g(y) = (\pi_1 g(y), \dots, \pi_n g(y)) = (f_1(y), \dots, f_n(y)) = f(y),$$

so $g = f$.

Since a unique mediating morphism exists for every candidate cone, $(\mathcal{M}, \varphi^t)$ with the projections π_j satisfies the universal property of finite products in **Dyn**. $\square$

Lemma D.2 (Coproducts). *For any family $(\mathcal{M}_i, \varphi_i^t)_{i \in I}$, the disjoint union*

$$\left(\bigsqcup_{i \in I} \mathcal{M}_i, \varphi^t(x) = \varphi_i^t(x) \text{ if } x \in \mathcal{M}_i \right)$$

*with the canonical injections $\iota_i : \mathcal{M}_i \hookrightarrow \bigsqcup_j \mathcal{M}_j$ is the coproduct in **Dyn**.*

Proof. Disjoint unions of compact manifolds are compact manifolds, the flow is C^k on each component, and the injections are equivariant by construction. The universal property follows from the universal property of disjoint unions in smooth manifolds. $\square$

Lemma D.3 (Regular equalisers). *Let $f, g : (\mathcal{M}, \varphi^t) \rightarrow (\mathcal{N}, \psi^t)$ be morphisms. Assume f and g are transverse to the diagonal $\Delta_{\mathcal{N}} \subset \mathcal{N} \times \mathcal{N}$. Then*

$$E := \{x \in \mathcal{M} \mid f(x) = g(x)\}$$

*is an embedded C^k sub-manifold, invariant under φ^t . The inclusion $i : E \hookrightarrow \mathcal{M}$ with the restricted flow $\varphi^t|_E$ realises the equaliser of (f, g) in **Dyn**.*

546 *Proof.* Transversality implies E is an embedded sub-manifold. Equivariance of f and g gives
 548 $f(\varphi^t x) = \psi^t f(x) = \psi^t g(x) = g(\varphi^t x)$, so E is flow-invariant. The inclusion i is a morphism,
 and any morphism $h : (\mathcal{K}, \chi^t) \rightarrow (\mathcal{M}, \varphi^t)$ with $fh = gh$ factors uniquely through i by the usual
 smooth-manifold argument. $\square$

550 *Remark D.3* (Limitations of completeness). The category of C^k manifolds is neither complete nor
 cocomplete; an infinite product need not be a manifold, and general coequalisers (or non-regular
 552 equalisers) can fail to carry a smooth Hausdorff structure. Consequently **Dyn** is *finitely* complete
 and has coproducts, but it does not possess arbitrary small limits or colimits.

554 **Lemma D.4** (Limits and colimits). **Dyn** has all small limits and colimits; they are inherited
 point-wise from the category of C^k manifolds.

556 *Sketch.* Products exist because finite products of compact manifolds are compact manifolds; the
 flow acts component-wise. Equalisers are submanifolds defined by equalising maps that commute
 558 with the flow. Dually for colimits. $\square$

The remainder of this section builds functors *out of Dyn* and uses natural transformations and
 560 comonads to formalise local linearisation and its residual correction.

D.2 Local linearisation as a natural transformation

562 Write $\Pi_d : \mathbf{Dyn} \rightarrow \mathbf{Dyn}$ for the functor that forgets all ≥ 2 -jet data. Weighted least squares at
 each anchor x induces the component

$$\eta_{(\mathcal{M}, \varphi^t)}(x) := B_x : F(\mathcal{M}, \varphi^t) \longrightarrow F \circ \Pi_d(\mathcal{M}, \varphi^t),$$

564 yielding a natural transformation $\eta : F \Rightarrow F \circ \Pi_d$.

D.3 Residual frame field as a cofree coalgebra

566 Define a comonad $\mathbf{G} : (\mathbf{Vect}, \otimes) \rightarrow (\mathbf{Vect}, \otimes)$ by $\mathbf{G}(V) = V \oplus (\text{span } Q_x)$. The map $x \mapsto (B_x, Q_x)$
 endows each fibre with the structure of a $\mathbf{G}$ -coalgebra, cofree under the variance-fraction constraint
 568 ρ . In string-diagram notation:

$$\begin{array}{ccc} F(\mathcal{M}, \varphi^t) & \xrightarrow{\quad \Delta \quad} & \mathbf{G}F(\mathcal{M}, \varphi^t) \\ \eta \downarrow & & \downarrow \mathbf{G}\eta \\ F \circ \Pi_d(\mathcal{M}, \varphi^t) & \xrightarrow{\quad \mathbf{G}\Delta \quad} & \mathbf{G}F \circ \Pi_d(\mathcal{M}, \varphi^t) \end{array}$$

The dashed arrows are the Rose augmentation; commutativity is enforced by letting $\mathbf{G}$ be *cofree*
 570 with respect to ρ .